

Quiz 11

Name: _____

Show all work clearly. Give exact answers unless otherwise stated.

1. (a) Convert the point

$$\left(4, \frac{\pi}{6}\right)$$

from polar form to rectangular form.

$$x = r \cos \theta = 4 \cdot \cos \frac{\pi}{6} = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$$

$$y = r \sin \theta = 4 \sin \frac{\pi}{6} = 4 \cdot \frac{1}{2} = 2$$

$$(2\sqrt{3}, 2)$$

- (b) Convert the point

$$(-2, 2\sqrt{3})$$

from rectangular form to polar form, where $r > 0$ and $0 \leq \theta < 2\pi$.

$$r = \sqrt{x^2 + y^2} = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$

$$\tan \theta = \frac{y}{x} = \frac{2\sqrt{3}}{-2} = -\sqrt{3}$$

$$\text{so } \theta = \frac{2\pi}{3}$$

$$(4, \frac{2\pi}{3})$$

2. Convert the polar equation

$$r = 6 \sin \theta$$

to rectangular form, and identify the curve.

$$r \cdot r = 6 \cdot r \cdot \sin \theta$$

$$x^2 + y^2 = 6y$$

$$x^2 + y^2 - 6y = 0$$

$$x^2 + (y-3)^2 = 9$$

circle, at $(0, 3)$, radius 3